

Curvature Cones and the Differentiable Sphere Theorem

Abstract

These notes explain the curvature-cone proof of the differentiable sphere theorem. The main construction starts with the weak PIC2 cone C_{PIC2} , deforms it into a family of transversally invariant cones, and uses that family to build a pinching set in Hamilton's sense. Hamilton's convergence theorem then gives a round limit. We also prove Berger's curvature estimate and use it to pass from pointwise quarter-pinching to PIC2. The final section gives Brendle's stronger PIC1 convergence theorem. Throughout, the notation C_{PIC2} is used consistently for the base weak-PIC2 cone; the deformed cones are denoted by $\mathcal{C}(s)$.

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1 Preliminaries

1.1 Historical background

The classical topological sphere theorem is the following.

Theorem 1.1 (Berger–Klingenberg topological sphere theorem [1, 9]). *Let (M^n, g) be compact and simply connected, with $n \geq 3$. If, after one global rescaling,*

$$\frac{1}{4} < K_g(\pi) \leq 1$$

for every two-plane $\pi \subset T_p M$ and every $p \in M$, then M is homeomorphic to $\mathbb{S}^n$.

The proof belongs to comparison geometry. A decisive input is Klingenberg's injectivity-radius estimate, which turns the curvature pinching into sufficiently strong global control of geodesics. The constant $1/4$ is optimal: the compact rank-one symmetric spaces have sectional curvatures in $[1/4, 1]$ after normalization.

The differentiable problem is subtler because a topological sphere need not carry the standard smooth structure. Ricci flow supplied the mechanism for deforming a suitable initial metric to a round one. Some milestones are:

- Hamilton proved that a compact three-manifold with $\text{Ric} > 0$ evolves, after normalization, to constant curvature [7].
- In dimension four, Hamilton obtained convergence under positive curvature operator [8].
- Böhm and Wilking extended the positive-curvature-operator result to every dimension [3].
- Brendle and Schoen proved the optimal differentiable quarter-pinched sphere theorem using the PIC2 cone [5].
- Brendle proved the stronger convergence theorem under the PIC1 condition [4].

The conclusion in the non-simply-connected case is naturally stated in terms of a *spherical space form*, namely a quotient $\mathbb{S}^n/\Gamma$, where $\Gamma < O(n+1)$ is finite and acts freely. If M is simply connected, this reduces to $M \cong \mathbb{S}^n$.

1.2 Curvature conditions and their basic properties

Let V be an n -dimensional Euclidean vector space, $n \geq 4$, and let $\mathcal{C}_B(V)$ denote the vector space of algebraic curvature tensors on V . Our sign convention is chosen so that the tensor

$$I(x, y, z, w) = \langle x, z \rangle \langle y, w \rangle - \langle x, w \rangle \langle y, z \rangle$$

has constant sectional curvature 1.

Definition 1.2 (PIC and WPIC). An algebraic curvature tensor R has *positive isotropic curvature* (PIC) if

$$R_{1313} + R_{1414} + R_{2323} + R_{2424} - 2R_{1234} > 0$$

for every orthonormal four-frame e . It has *weakly positive isotropic curvature* (WPIC) if the same expression is nonnegative.

Equivalently, after complex-bilinearly extending the metric and curvature, PIC means

$$R(\zeta, \eta, \bar{\zeta}, \bar{\eta}) > 0$$

whenever ζ, η span a totally isotropic complex two-plane, that is,

$$g(\zeta, \zeta) = g(\zeta, \eta) = g(\eta, \eta) = 0.$$

Replacing $>$ by $\geq$ gives WPIC.

Definition 1.3 (PIC1 and PIC2). Let R be the curvature tensor of (M, g) .

(i) R is PIC1 if $M \times \mathbb{R}$ has PIC, or equivalently

$$R_{1313} + \lambda^2 R_{1414} + R_{2323} + \lambda^2 R_{2424} - 2\lambda R_{1234} > 0 \quad (1.1)$$

for every orthonormal four-frame and every $\lambda \in [0, 1]$.

(ii) R is PIC2 if $M \times \mathbb{R}^2$ has PIC, or equivalently

$$R_{1313} + \lambda^2 R_{1414} + \mu^2 R_{2323} + \lambda^2 \mu^2 R_{2424} - 2\lambda \mu R_{1234} > 0 \quad (1.2)$$

for every orthonormal four-frame and every $\lambda, \mu \in [0, 1]$.

The corresponding non-strict conditions are denoted WPIC1 and WPIC2. We write C_{PIC1} and C_{PIC2} for the closed cones of WPIC1 and WPIC2 tensors, respectively. Thus PIC1 and PIC2 are the interiors of these cones.

The complex formulation of PIC2 is particularly simple:

$$R(\zeta, \eta, \bar{\zeta}, \bar{\eta}) \geq 0 \quad \text{for all } \zeta, \eta \in V \otimes \mathbb{C}.$$

For PIC1, the same inequality is required on complex two-planes for which the restriction of the complex-bilinear metric is degenerate. In the four-frame characterization above, this is the condition $(1 - \lambda^2)(1 - \mu^2) = 0$, so at least one of λ, μ equals 1.

The main weak conditions are related as follows:

$$\begin{array}{ccccc} \text{nonnegative curvature operator} & \implies & \text{WPIC2} & \implies & K \geq 0 \\ \Downarrow & & \Downarrow & & \Downarrow \\ \text{two-nonnegative curvature operator} & \implies & \text{WPIC1} & \implies & \text{Ric} \geq 0 \\ & & \Downarrow & & \Downarrow \\ & & \text{WPIC} & \implies & \text{scal} \geq 0. \end{array}$$

For example, WPIC2 implies $K \geq 0$ by setting $\lambda = \mu = 0$ in the defining inequality. WPIC1 implies $\text{Ric} \geq 0$ by letting $\lambda = 0$ and summing the resulting inequalities. Nonnegative curvature operator implies WPIC2 because, with

$$\varphi = e_1 \wedge e_3 + \lambda \mu e_4 \wedge e_2, \quad \psi = \lambda e_1 \wedge e_4 + \mu e_2 \wedge e_3,$$

the first Bianchi identity gives

$$R(\varphi, \varphi) + R(\psi, \psi) = R_{1313} + \lambda^2 R_{1414} + \mu^2 R_{2323} + \lambda^2 \mu^2 R_{2424} - 2\lambda \mu R_{1234}.$$

The closed cones C_{PIC} , C_{PIC1} , and C_{PIC2} are convex and $O(n)$ -invariant. They are also preserved by the Hamilton ODE and hence, by Hamilton's tensor maximum principle, by Ricci flow [6, Chapters 5 and 7]. The preservation of C_{PIC1} and C_{PIC2} may also be seen by applying preservation of WPIC to the products $M \times \mathbb{R}$ and $M \times \mathbb{R}^2$.

1.3 Pinching sets and Hamilton's convergence theorem

For $\delta \in (0, 1)$, an algebraic curvature tensor is *strictly δ -pinched* if

$$0 < \delta K_R(\pi_1) < K_R(\pi_2)$$

for every pair of two-planes π_1, π_2 . It is *weakly δ -pinched* if

$$0 \leq \delta K_R(\pi_1) \leq K_R(\pi_2)$$

for every such pair.

The curvature part of the Ricci-flow evolution equation is the quadratic map $Q : \mathcal{C}_B(\mathbb{R}^n) \rightarrow \mathcal{C}_B(\mathbb{R}^n)$ given by

$$Q(R)_{ijkl} = \sum_{p,q} R_{ijpq} R_{klpq} + 2 \sum_{p,q} (R_{ipkq} R_{jplq} - R_{iplq} R_{jpkq}). \quad (1.3)$$

The ODE

$$\frac{d}{dt} R = Q(R) \quad (1.4)$$

is called the *Hamilton ODE*.

Definition 1.4 (Pinching set). A subset $F \subset \mathcal{C}_B(\mathbb{R}^n)$ is a *pinching set* if:

- (i) F is closed, convex, and $O(n)$ -invariant;

- (ii) F is invariant under (1.4);
- (iii) for every $\delta \in (0, 1)$, the set

$$\{R \in F : R \text{ is not weakly } \delta\text{-pinched}\}$$

is bounded.

The third condition says that the unbounded end of F becomes arbitrarily close to the ray $\mathbb{R}_+ I$. It is precisely the algebraic pinching needed in the following theorem.

Theorem 1.5 (Hamilton's convergence criterion [8]). *Let M^n be compact, $n \geq 3$, and let g_0 have positive scalar curvature. Suppose there is a pinching set $F \subset \mathcal{C}_B(\mathbb{R}^n)$ such that $\text{Rm}_{g_0}(p) \in F$ for every $p \in M$. If $g(t)$, $t \in [0, T)$, is the maximal Ricci flow starting at g_0 , then $T < \infty$ and*

$$\frac{1}{2(n-1)(T-t)} g(t)$$

converges in C^∞ to a metric of constant sectional curvature 1.

Thus, after positivity of the scalar curvature is known, the geometric convergence problem is reduced to constructing a pinching set containing the initial curvature tensors.

2 Proof of the Differentiable Sphere Theorem

Idea

The base cone C_{PIC2} is preserved by Ricci flow, but it is not itself a pinching set: it contains many non-round rays. We therefore deform it through a family $\mathcal{C}(s)$ of invariant cones. As $s \rightarrow \infty$, the cones become more and more pinched toward $\mathbb{R}_+ I$. Two continuation lemmas then let us move any compact set of strictly PIC2 tensors through the entire family. The smallest closed, convex, $O(n)$ - and ODE-invariant set containing the initial curvatures is consequently a pinching set.

2.1 Step I: construction and pinching of the cone family

For symmetric bilinear forms A, B , their Kulkarni–Nomizu product is

$$(A \wedge B)_{ijkl} = A_{ik}B_{jl} - A_{il}B_{jk} - A_{jk}B_{il} + A_{jl}B_{ik}.$$

For $a, b > 0$, define a linear map $\ell_{a,b} : \mathcal{C}_B(\mathbb{R}^n) \rightarrow \mathcal{C}_B(\mathbb{R}^n)$ by

$$\ell_{a,b}(R) = R + b \text{Ric}_R \wedge \text{id} + \frac{a-b}{n} \text{scal}_R \text{id} \wedge \text{id}. \quad (2.1)$$

The curvature decomposition

$$R = W + \frac{1}{n-2} \text{Ric}_R^\circ \wedge \text{id} + \frac{\text{scal}_R}{2n(n-1)} \text{id} \wedge \text{id}$$

shows that $\ell_{a,b}$ multiplies the Weyl, trace-free Ricci, and scalar summands by

$$1, \quad 1 + (n-2)b, \quad 1 + 2(n-1)a,$$

respectively. Hence $\ell_{a,b}$ is an isomorphism. In particular,

$$\ell_{a,b}(I) = (1 + 2(n-1)a)I. \quad (2.2)$$

The key Böhm–Wilking identity explains the particular parameters used below. If $\text{Ric}^\circ = \text{Ric} - \frac{1}{n} \text{scal id}$ and $(\text{Ric}^\circ)_{ij}^2 = \text{Ric}_{ik}^\circ \text{Ric}_{kj}^\circ$, set

$$D_{a,b}(R) = ((n-2)b^2 - 2(a-b)) \text{Ric}^\circ \wedge \text{Ric}^\circ + 2a \text{Ric} \wedge \text{Ric} + 2b^2 (\text{Ric}^\circ)^2 \wedge \text{id} \\ + \frac{nb^2(1-2b) - 2(a-b)(1-2b+nb^2)}{n(1+2(n-1)a)} |\text{Ric}^\circ|^2 \text{id} \wedge \text{id}. \quad (2.3)$$

The following result is just pure computation thus we don't give the proof, and one can refer to [6].

Proposition 2.1 (Böhm–Wilking identity). *For every algebraic curvature tensor R ,*

$$\ell_{a,b}^{-1}(Q(\ell_{a,b}(R))) = Q(R) + D_{a,b}(R). \quad (2.4)$$

Proof. Insert (2.1) into (1.3), separate the Weyl, trace-free Ricci, and scalar parts, and use the first Bianchi identity to combine the mixed terms. The pure R terms give $Q(R)$; the remaining terms are exactly (2.3). This is the useful point of the calculation: under $\ell_{a,b}$, the Hamilton vector field is conjugated to $Q + D_{a,b}$. $\square$

We record the algebraic cone-deformation statement used in the construction. A cone $C \subset \mathcal{C}_B(\mathbb{R}^n)$ is said to have property $(*)$ if it is closed, convex, and $O(n)$ -invariant, is invariant under the Hamilton ODE, consists of tensors with nonnegative sectional curvature, and contains every tensor with nonnegative curvature operator.

Lemma 2.2. *Suppose C has property $(*)$.*

(i) *If $0 < b \leq \frac{1}{2}$ and*

$$2a = \frac{2b + (n-2)b^2}{1 + (n-2)b^2}, \quad \delta = \frac{(n-2)b^2}{1 + (n-2)b^2},$$

then

$$\{\ell_{a,b}(R) : R \in C, \text{Ric}_R \geq \frac{\delta}{n} \text{scal}_R \text{id}\}$$

is transversally invariant under the Hamilton ODE away from the origin.

(ii) *If $a > \frac{1}{2}$, $b = \frac{1}{2}$, and*

$$\delta = 1 - \frac{4}{n-2+8a},$$

the same conclusion holds.

The verification of transversality is given in [6, Propositions 8.6 and 8.7].

The cone C_{PIC2} has property $(*)$: WPIC2 implies nonnegative sectional curvature; non-negative curvature operator implies WPIC2; and C_{PIC2} is closed, convex, $O(n)$ -invariant, and Hamilton-ODE invariant. Define, for $s > 0$,

$$2a(s) = \begin{cases} \frac{2s + (n-2)s^2}{1 + (n-2)s^2}, & 0 < s \leq \frac{1}{2}, \\ 2s, & s > \frac{1}{2}, \end{cases} \quad 2b(s) = \begin{cases} 2s, & 0 < s \leq \frac{1}{2}, \\ 1, & s > \frac{1}{2}, \end{cases} \quad (2.5)$$

$$\delta(s) = \begin{cases} 1 - \frac{1}{1 + (n-2)s^2}, & 0 < s \leq \frac{1}{2}, \\ 1 - \frac{4}{n-2+8s}, & s > \frac{1}{2}. \end{cases} \quad (2.6)$$

These functions agree at $s = \frac{1}{2}$ and are continuous. We now define the deformed cone

$$\mathcal{C}(s) = \left\{ \ell_{a(s),b(s)}(R) : R \in C_{\text{PIC2}}, \text{Ric}_R \geq \frac{\delta(s)}{n} \text{scal}_R \text{id} \right\}. \quad (2.7)$$

Notice that the base cone is always denoted C_{PIC2} ; $\mathcal{C}(s)$ is the new, parameter-dependent cone. Moreover, $a(s), b(s), \delta(s) \rightarrow 0$ as $s \downarrow 0$, so $\mathcal{C}(s)$ tends to C_{PIC2} .

The cones $\mathcal{C}(s)$ are closed, convex, $O(n)$ -invariant, contain I in their interiors, and are transversally invariant under the Hamilton ODE; see [6, Proposition 8.8(i)]. We only prove the pinching property needed below.

Proposition 2.3 (Pinching of the cone family). *If $s > \frac{1}{2}$, every $S \in \mathcal{C}(s)$ is weakly*

$$\frac{2s-1}{2s+n-1}\text{-pinched}.$$

Proof. Suppose $s > 1/2$ and write $S = \ell_{s,1/2}(R)$, where $R \in C_{\text{PIC2}}$ and $\text{Ric}_R \geq \frac{\delta(s)}{n} \text{scal}_R \text{id}$. Since R has nonnegative sectional curvature, it has nonnegative Ricci curvature. For every orthonormal two-frame (e_1, e_2) , (2.1) gives

$$S_{1212} = R_{1212} + \frac{1}{2}(\text{Ric}_{11} + \text{Ric}_{22}) + \frac{2s-1}{n} \text{scal}_R. \quad (2.8)$$

Therefore

$$S_{1212} \geq \frac{2s-1}{n} \text{scal}_R.$$

On the other hand, nonnegative sectional curvature gives $R_{1212} \leq \frac{1}{2}(\text{Ric}_{11} + \text{Ric}_{22})$, and $\text{Ric}_{11} + \text{Ric}_{22} \leq \text{scal}_R$. Hence

$$S_{1212} \leq \frac{2s+n-1}{n} \text{scal}_R.$$

The ratio of the lower and upper bounds is $(2s-1)/(2s+n-1)$, proving the claim. $\square$

2.2 Step II: the two continuation lemmas

The next lemma moves an invariant set a uniform positive distance along the cone family. For a closed convex set C and $R \in C$, $T_R C$ denotes its tangent cone at R .

Lemma 2.4 (Continuation along the cone family; Brendle, Lemma 8.9). *Fix a compact interval $[\alpha, \beta] \subset (0, \infty)$. There exists $\varepsilon = \varepsilon(\alpha, \beta, n) > 0$ with the following property. Suppose $F \subset \mathcal{C}_B(\mathbb{R}^n)$ is closed and invariant under the Hamilton ODE, and for some $s \in [\alpha, \beta]$ and $h > 0$,*

$$F \subset \{R : R + hI \in \mathcal{C}(s)\}.$$

Then

$$\tilde{F} := \{R \in F : R + 2hI \in \mathcal{C}(s + \varepsilon)\}$$

is also invariant under the Hamilton ODE. Moreover,

$$\{R \in F : \text{scal}_R \leq h\} \subset \tilde{F}.$$

For the proof, see [6, Lemma 8.9].

Lemma 2.5 (Construction of a pinching set; Brendle, Proposition 8.10). *Let $K \subset \mathcal{C}_B(\mathbb{R}^n)$ be compact, and let F be the smallest subset of $\mathcal{C}_B(\mathbb{R}^n)$ that contains K and is closed, convex, $O(n)$ -invariant, and invariant under the Hamilton ODE. If*

$$F \subset \{R : R + h_0 I \in \mathcal{C}(s_0)\}$$

for some $s_0, h_0 > 0$, then F is a pinching set.

Proof. Let

$$Y = \{s > 0 : \text{there exists } h > 0 \text{ with } F \subset \{R : R + hI \in \mathcal{C}(s)\}\}.$$

By assumption, Y is nonempty. Set $\sigma = \sup Y$ and choose $s_j \in Y$ with $s_j \rightarrow \sigma$. For each j , choose $h_j > 0$ such that

$$F \subset \{R : R + h_j I \in \mathcal{C}(s_j)\}.$$

Increasing h_j if necessary, we may assume

$$h_j > \sup_{R \in K} \text{scal}_R. \quad (2.9)$$

We claim that $\sigma = \infty$. Otherwise the s_j lie in a compact subinterval of $(0, \infty)$. By [lemma 2.4](#), one number $\varepsilon > 0$ works for every j , and

$$F_j := \{R \in F : R + 2h_j I \in \mathcal{C}(s_j + \varepsilon)\}$$

is Hamilton-ODE invariant. It is also closed, convex, and $O(n)$ -invariant. By (2.9) and the final assertion of [lemma 2.4](#), $K \subset F_j$. The minimality of F therefore implies $F \subset F_j$. Hence $s_j + \varepsilon \in Y$, contradicting $s_j \rightarrow \sup Y$. Thus $\sup Y = \infty$.

It remains to verify the last condition in the definition of a pinching set. Fix $\delta \in (0, 1)$. By [Proposition 2.3](#), choose s so large that every tensor in $\mathcal{C}(s)$ is weakly δ' -pinched for some $\delta' > \delta$. Since $s \in Y$, there is $h > 0$ such that $R + hI \in \mathcal{C}(s)$ for every $R \in F$. Write $S = R + hI$. If R is not weakly δ -pinched, then either $K_{\min}(R) < 0$, or $K_{\min}(R) < \delta K_{\max}(R)$. In the first case,

$$\delta' K_{\max}(S) \leq K_{\min}(S) < h;$$

in the second,

$$(\delta' - \delta)K_{\max}(S) < (1 - \delta)h.$$

Thus all sectional curvatures of S , and hence of $R = S - hI$, are uniformly bounded. Polarization bounds every component of an algebraic curvature tensor in terms of its sectional curvatures, so the set of such R is bounded in $\mathcal{C}_B(\mathbb{R}^n)$. Therefore F is a pinching set. $\square$

2.3 Step III: PIC2 implies a spherical space form

Theorem 2.6 (PIC2 differentiable sphere theorem). *Let (M^n, g_0) be compact, $n \geq 4$, and suppose g_0 is PIC2. Then the maximal Ricci flow $g(t)$ becomes round after normalization:*

$$\frac{1}{2(n-1)(T-t)}g(t) \longrightarrow g_\infty \quad \text{in } C^\infty,$$

where g_∞ has constant sectional curvature 1. Consequently, M is diffeomorphic to a spherical space form. If M is simply connected, then M is diffeomorphic to $\mathbb{S}^n$.

Proof. The set

$$K = \{\text{Rm}_{g_0}(p) : p \in M\} \subset \mathcal{C}_B(\mathbb{R}^n)$$

is compact and lies in $\text{int } C_{\text{PIC2}}$. Since $\mathcal{C}(s)$ tends to C_{PIC2} as $s \downarrow 0$, compactness gives an $s_0 > 0$ such that $K \subset \text{int } \mathcal{C}(s_0)$. Let F be the smallest closed, convex, $O(n)$ -invariant and Hamilton-ODE-invariant set containing K . Since $\mathcal{C}(s_0)$ has all of these properties, $F \subset \mathcal{C}(s_0)$. Since $I \in \mathcal{C}(s_0)$ and $\mathcal{C}(s_0)$ is a convex cone, $F + hI \subset \mathcal{C}(s_0)$ for every $h \geq 0$. Thus the hypothesis of [lemma 2.5](#) holds, and F is a pinching set.

PIC2 implies positive sectional curvature by taking $\lambda = \mu = 0$ in (1.2), so $\text{scal}_{g_0} > 0$. Hamilton's convergence theorem, [theorem 1.5](#), now gives the stated round limit. A compact constant-positive-curvature manifold is a spherical space form, and the normalized Ricci flow is a smooth path of metrics on the fixed manifold M , so the diffeomorphism conclusion follows. $\square$

2.4 Step IV: quarter-pinching implies PIC2

We first prove the Berger estimate used in the final implication.

Proposition 2.7 (Berger's curvature estimate [2]). *Suppose that, at a point p , every sectional curvature lies in $[\kappa_{\min}, \kappa_{\max}]$. Then, for every orthonormal four-frame,*

$$|R_{1234}| \leq \frac{2}{3}(\kappa_{\max} - \kappa_{\min}).$$

Proof. For brevity write $\mathcal{R}(x, y) := R(x, y, x, y)$. Multilinearity gives

$$\begin{aligned} A &:= \mathcal{R}(e_1 + e_3, e_2 + e_4) - \mathcal{R}(e_1 + e_3, e_2 - e_4) \\ &\quad - \mathcal{R}(e_1 - e_3, e_2 + e_4) + \mathcal{R}(e_1 - e_3, e_2 - e_4) \\ &= 8R_{1234} + 8R_{1432}, \\ B &:= \mathcal{R}(e_1 + e_4, e_2 + e_3) - \mathcal{R}(e_1 + e_4, e_2 - e_3) \\ &\quad - \mathcal{R}(e_1 - e_4, e_2 + e_3) + \mathcal{R}(e_1 - e_4, e_2 - e_3) \\ &= 8R_{1243} + 8R_{1342}. \end{aligned}$$

Subtracting and using the first Bianchi identity yields

$$A - B = 24R_{1234}. \quad (2.10)$$

Each pair of vectors appearing above is orthogonal and both vectors have length $\sqrt{2}$. Hence every term $\mathcal{R}(x, y)$ lies between $4\kappa_{\min}$ and $4\kappa_{\max}$. In the signed sum $A - B$, four terms have positive sign and four have negative sign. Therefore

$$24R_{1234} \leq 16(\kappa_{\max} - \kappa_{\min}).$$

Applying the same estimate after replacing e_4 by $-e_4$ gives the corresponding lower bound. Division by 24 proves the result. $\square$

Corollary 2.8 (Pointwise quarter-pinching implies PIC2). *If a Riemannian manifold is strictly quarter-pinched in the pointwise sense, then it is PIC2.*

Proof. Fix a point and write $K_{\min}$ and $K_{\max}$ for the smallest and largest sectional curvatures there. Strict pointwise quarter-pinching means

$$0 < K_{\min}, \quad K_{\max} < 4K_{\min}.$$

By Proposition 2.7,

$$|R_{1234}| \leq \frac{2}{3}(K_{\max} - K_{\min}) < 2K_{\min}.$$

Consequently, for every $\lambda, \mu \in [0, 1]$,

$$\begin{aligned} R_{1313} + \lambda^2 R_{1414} + \mu^2 R_{2323} + \lambda^2 \mu^2 R_{2424} - 2\lambda\mu R_{1234} &\geq (1 + \lambda^2 + \mu^2 + \lambda^2 \mu^2)K_{\min} - 2\lambda\mu |R_{1234}| \\ &> (1 + \lambda^2 + \mu^2 + \lambda^2 \mu^2 - 4\lambda\mu)K_{\min} \\ &= ((1 - \lambda\mu)^2 + (\lambda - \mu)^2)K_{\min} \geq 0. \end{aligned}$$

The first inequality is strict enough to give positivity even when $\lambda = \mu = 1$, where the final polynomial vanishes. Thus the full curvature expression is positive, which is exactly PIC2. $\square$

Combining corollary 2.8 and theorem 2.6 gives the differentiable quarter-pinched sphere theorem of Brendle–Schoen: every compact strictly pointwise quarter-pinched manifold is diffeomorphic to a spherical space form.

3 The Improved PIC1 Convergence Theorem

PIC1 is weaker than PIC2, so a compact subset of $\text{int } C_{\text{PIC1}}$ need not start inside the cone family $\mathcal{C}(s)$. The bridge is an auxiliary Hamilton-invariant set corresponding to positive isotropic curvature on a product with a round two-sphere.

For $N > 0$, define $G_N \subset \mathcal{C}_B(\mathbb{R}^n)$ by the inequalities

$$R_{1313} + \lambda^2 R_{1414} + \mu^2 R_{2323} + \lambda^2 \mu^2 R_{2424} - 2\lambda\mu R_{1234} + N(1 - \lambda^2)(1 - \mu^2) \geq 0 \quad (3.1)$$

for every orthonormal four-frame and every $\lambda, \mu \in [0, 1]$. This is equivalent to WPIC on the product of V with a round two-sphere of sectional curvature N . We write $G = G_1$.

Proposition 3.1 (Properties of the intermediate set). *The set G_N is closed, convex, $O(n)$ -invariant, and invariant under the Hamilton ODE. Moreover,*

$$C_{\text{PIC2}} \subset G_N \subset C_{\text{PIC1}}, \quad G_N + NI \subset C_{\text{PIC2}}. \quad (3.2)$$

Proof. Closedness, convexity, and $O(n)$ -invariance follow directly from (3.1). The first inclusion in (3.2) is immediate because the added term is nonnegative. For PIC1, at least one of λ, μ equals 1, so the added term vanishes; this proves $G_N \subset C_{\text{PIC1}}$. Adding NI adds

$$N(1 + \lambda^2)(1 + \mu^2)$$

to the curvature expression. Together with (3.1), this is nonnegative, proving $G_N + NI \subset C_{\text{PIC2}}$.

Hamilton-ODE invariance is proved in [6, Proposition 7.28]; the statement for general $N > 0$ follows by scaling from the case $N = 1$. $\square$

The next deformation result is the key new ingredient in the PIC1 argument.

Proposition 3.2 (Invariant deformation of G_N). *Suppose*

$$2a = 2b + (n - 2)b^2, \quad 0 < b \leq \frac{\sqrt{2n(n - 2) + 4} - 2}{n(n - 2)}. \quad (3.3)$$

Then $\ell_{a,b}(G_N)$ is invariant under the Hamilton ODE.

This is [6, Proposition 8.15]; the case of G_N follows from the stated case G_1 by scaling.

We also use the following direct consequence of the definitions and the continuity of the cone family.

Lemma 3.3 (Entry into the PIC2 cone family). *For $a, b > 0$ satisfying (3.3), there exists $s_0 > 0$ such that*

$$\ell_{a,b}(C_{\text{PIC1}}) \subset \mathcal{C}(s_0).$$

Proof. The term $b \text{Ric} \wedge \text{id}$ in (2.1) turns the one-flat-factor PIC1 inequalities into the two-flat-factor inequalities, while $a - b = \frac{n-2}{2}b^2 > 0$ supplies the required scalar margin. The same calculation gives a uniform lower bound $\text{Ric} \geq \frac{\delta(s_0)}{n} \text{scal id}$ after applying $\ell_{a(s_0), b(s_0)}^{-1}$. Choosing s_0 sufficiently small makes $a(s_0), b(s_0), \delta(s_0)$ fit these strict margins. Since all statements are homogeneous and $O(n)$ -invariant, it is enough to check them on a compact normalized slice of C_{PIC1} , where the required s_0 is uniform. $\square$

Theorem 3.4 (Brendle's PIC1 convergence theorem [4]). *Let (M^n, g_0) be compact, $n \geq 4$, and suppose g_0 is PIC1. Then the normalized Ricci flow converges in C^∞ to a metric of constant sectional curvature 1. In particular, M is diffeomorphic to a spherical space form.*

Proof. Let

$$K = \{\text{Rm}_{g_0}(p) : p \in M\}.$$

This is a compact subset of $\text{int } C_{\text{PIC1}}$. The PIC1 characterization says

$$R_{1313} + \lambda^2 R_{1414} + \mu^2 R_{2323} + \lambda^2 \mu^2 R_{2424} - 2\lambda\mu R_{1234} > 0$$

whenever $(1 - \lambda^2)(1 - \mu^2) = 0$. By compactness, there exists $N > 0$ such that

$$R_{1313} + \lambda^2 R_{1414} + \mu^2 R_{2323} + \lambda^2 \mu^2 R_{2424} - 2\lambda\mu R_{1234} + N(1 - \lambda^2)(1 - \mu^2) > 0$$

for every $R \in K$, every orthonormal four-frame, and all $\lambda, \mu \in [0, 1]$. Thus $K \subset \text{int } G_N$.

Choose a, b as in (3.3), with $b > 0$ sufficiently small. Since $\ell_{a,b} \rightarrow \text{id}$ as $b \downarrow 0$, compactness gives $K \subset \text{int } \ell_{a,b}(G_N)$. Let F be the smallest closed, convex, $O(n)$ -invariant and Hamilton-ODE-invariant set containing K . By Proposition 3.2,

$$F \subset \ell_{a,b}(G_N).$$

For every $R \in G_N$, (3.2), (2.2), and lemma 3.3 give

$$\begin{aligned} \ell_{a,b}(R) + N(1 + 2(n-1)a)I &= \ell_{a,b}(R + NI) \\ &\in \ell_{a,b}(C_{\text{PIC2}}) \subset \ell_{a,b}(C_{\text{PIC1}}) \subset \mathcal{C}(s_0). \end{aligned}$$

Hence

$$F \subset \{S : S + N(1 + 2(n-1)a)I \in \mathcal{C}(s_0)\}.$$

By lemma 2.5, F is a pinching set. Strict PIC1 implies positive Ricci, hence positive scalar curvature. Hamilton's convergence criterion, theorem 1.5, now yields the round limit and the spherical-space-form conclusion. $\square$

The logical structure of the improved argument is therefore

$$\begin{aligned} \text{PIC1} &\implies K \Subset \text{int } G_N \implies K \subset \ell_{a,b}(G_N) \\ &\implies F + hI \subset \mathcal{C}(s_0) \implies F \text{ is a pinching set} \implies \text{round convergence.} \end{aligned}$$

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